Contact

www.linkedin.com/in/hariharan-babu-hb (LinkedIn)

Top Skills

Agentic AI & LLM Development
Machine Learning & Deep Learning
Full Stack Web Development (MERN Stack)

Languages

English (Full Professional)
Tamil (Native or Bilingual)
Telugu (Native or Bilingual)

Honors-Awards

First Class – Bachelor of Engineering
Dean’s List – Master of Science

Hariharan Babu

Aspiring Data Engineer | ETL/ELT Pipelines | Big Data (PySpark, Kafka, Airflow) | Cloud (AWS, GCP, Azure) | Data Warehousing (Snowflake, BigQuery, Redshift)
Dallas, Texas, United States

Summary

I am currently pursuing a Master’s in Data Science with a strong passion for Data Engineering. I focus on developing end-to-end data architectures using technologies such as Snowflake, Databricks, and Delta Lake to support unified analytics. My experience includes building ETL/ELT pipelines, implementing data quality and governance frameworks (Great Expectations, OpenLineage), and delivering real-time insights through Tableau, Power BI tools and API-driven dashboards.I enjoy working at the intersection of data engineering, cloud platforms, and distributed systems, leveraging Python (PySpark), SQL, and streaming technologies to create performance-optimized data solutions.Some of the technologies I most frequently work with are: ☁️# Cloud: AWS (S3, EC2, EMR, Glue, Lambda, EKS, IAM, RDS, Redshift, CloudWatch), GCP (BigQuery, Dataflow), Azure (Azure Data Factory, Azure Databricks) Databases & Warehousing: SQL (Advanced Joins, CTEs, Window Functions, Query Optimization), Amazon Redshift, Google BigQuery, Snowflake, Amazon RDS, Hadoop (HDFS)## Data Engineering / ETL / Big Data: Apache Spark (PySpark), Kafka, MapReduce, Pig, Sqoop, dbt (models, tests, snapshots), ETL/ELT Pipelines, Batch & Stream Processing, Data Modeling (Star & Snowflake Schema) BI / Analytics / Visualization: Tableau, Looker, Microsoft Excel (Advanced, VBA, Solver), Google Sheets, Python (Matplotlib, Seaborn), Google Analytics Machine Learning & AI: Scikit-Learn, TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Transformers, Hugging Face, Embeddings, Tokenization, LLM Fine-Tuning, RAG, GANs## Workflow / DevOps: Apache Airflow, Terraform (Infrastructure as Code), CI/CD Pipelines, GitHub, AWS CloudWatchI thrive in collaborative, high-impact environments where engineering excellence meets innovation. My goal is to build data platforms that enable smarter, faster, and data-driven decision-making. Currently exploring Data Engineering opportunities, where I can contribute to next-generation cloud data platforms. Let’s

connect if you’d like to discuss data engineering, cloud strategy, or AI-driven data solutions.Skills- Python- SQL- PySpark- Gen AI- RAG (Retrieval-Augmented Generation)- LangChain- LangGraph- AI Agents & Agentic AI- Machine Learning- Data Engineering- ETL Processes- Cloud Platforms (AWS, GCP, Azure)

Experience

University of North Texas
Graduate Student Research Assistant
January 2025 - Present (1 year 2 months)
Dallas, TX

Advanced statistical learning and large-scale data research by engineering scalable ML pipelines and distributed workflows, strengthening model robustness and improving experimental reproducibility by 32%.
Designed and evaluated agentic AI systems capable of autonomous reasoning and multi-step orchestration, reducing manual research intervention by 46% while enhancing execution reliability.
Optimized big data processing for semi-structured and high-dimensional datasets through feature abstraction and dimensionality strategies, increasing downstream predictive stability by 28%.
Led end-to-end career readiness initiatives for 600+ graduate students, architecting workshops on résumé strategy, LinkedIn positioning, interview performance, and strategic networking to elevate market competitiveness.
Delivered structured mentorship, faculty collaboration, and performance-based feedback frameworks, strengthening professional branding quality and improving interview preparedness across student cohorts.

Skills: Python (Programming Language) · Machine Learning · Agentic AI Systems · Large Language Models (LLMs) · Deep Learning · Distributed Computing · Statistical Modeling · Data Engineering · Model Validation · Career Coaching · Resume & LinkedIn Strategy · Interview Preparation · Executive Communication · Workshop Facilitation · Stakeholder Collaboration

Newgen KnowledgeWorks
Data Engineer
June 2023 - July 2024 (1 year 2 months)
Chennai, Tamil Nadu, India

Designed and deployed a serverless, event-driven ETL pipeline on AWS, leveraging S3 event notifications and Lambda functions for automated metadata extraction and processing.
☁#Developed robust data transformation workflows using AWS Glue Jobs and Glue DataBrew, cleansing and normalizing nested JSON datasets to optimize analytical readiness and query performance.
Implemented a scalable archival solution migrating unstructured data (XML, JSON, PDFs, documents, images) from Box to S3, ensuring reliability and traceability with Python-based pipelines.
Engineered a fault-tolerant SQS and DLQ processing framework, incorporating threshold-based retry mechanisms and re-routing for error handling and exhaustive processing.
Secured data and enhanced observability, encrypting files using AES-256, indexing metadata into Amazon OpenSearch, and monitoring pipelines with CloudWatch Events and SNS alerts.

Skills: Python (Programming Language) · AWS · Amazon S3 · AWS Lambda · AWS Glue · Glue DataBrew · Amazon SQS · Dead Letter Queue (DLQ) · Event-Driven Architecture · ETL Pipelines · Unstructured Data Processing · Data Transformation · Cloud Monitoring · CloudWatch · SNS Notifications · Amazon OpenSearch · Data Security · AES-256 Encryption · Scalable System Design · Fault-Tolerant Systems

iHUB DivyaSampark @ IIT Roorkee
Data Science Intern
April 2023 - May 2023 (2 months)

Executed the complete data science pipeline, encompassing data ingestion, preprocessing, exploratory data analysis, feature engineering, predictive modeling, and performance evaluation on real-world datasets.
Engineered and validated machine learning models using Python, leveraging Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn to derive statistically sound insights.
##Collaborated with domain mentors to deliver industry-grade analytical solutions, translating business requirements into scalable data-driven models.
Applied rigorous evaluation techniques, including cross-validation and metric optimization, to ensure model robustness and generalization.
Acquired practical exposure to model deployment concepts, emphasizing reproducibility, interpretability, and operational readiness.

Skills: Python (Programming Language) · Data Science · Data Ingestion · Data Preprocessing · Exploratory Data Analysis (EDA) · Feature Engineering · Predictive Modeling · Machine Learning · Model Evaluation · Cross-Validation · Performance Metrics · Pandas · NumPy · Scikit-learn · Data Visualization · Matplotlib · Seaborn · Model Deployment Concepts · Reproducible Machine Learning · Model Interpretability · Statistical Analysis.

National Institute of Technology Karnataka

Machine Learning Intern

January 2023 - March 2023 (3 months)

Designed and implemented supervised machine learning models to analyze and predict outcomes from high-dimensional datasets using Python-based ML frameworks.

Conducted systematic data preprocessing, including outlier treatment, feature scaling, and dimensionality reduction, to enhance model accuracy and stability.

Benchmarked multiple algorithms—Linear & Logistic Regression, Decision Trees, KNN—and optimized performance through hyperparameter tuning.

Interpreted model outputs using quantitative metrics, enabling data-backed decision-making and algorithm selection.

##Strengthened applied understanding of algorithmic trade-offs, bias-variance considerations, and predictive performance optimization.

Skills: Python (Programming Language) · Machine Learning · Supervised Learning · Data Preprocessing · Feature Scaling · Dimensionality Reduction · Regression Analysis · Classification Algorithms · Decision Trees · K-Nearest Neighbors (KNN) · Hyperparameter Tuning · Model Evaluation Metrics · Algorithm Benchmarking · Bias–Variance Tradeoff · Predictive Modeling · Data Analysis

IEEE

Machine Learning Intern

May 2022 - July 2022 (3 months)

Developed a strong theoretical and practical foundation in supervised and unsupervised learning, including regression, classification, and clustering paradigms.

Implemented machine learning algorithms using Python, focusing on mathematical intuition, convergence behavior, and model interpretability.

Processed unstructured and semi-structured datasets, applying feature extraction and normalization techniques to prepare data for modeling.

##Conducted experimental analysis to compare algorithmic performance across datasets and learning strategies.
Cultivated analytical rigor and structured problem-solving skills through hands-on experimentation and guided research-oriented tasks.

Skills: Python (Programming Language) · Machine Learning · Agentic AI & LLM Development · Deep Learning · Computer Vision · Mathematical Modelling · Tableau · Optimization Techniques.

Education

University of North Texas
Master's degree, Data Science · (August 2024 - May 2026)

Anna University Chennai
Bachelor's degree, Computer Science · (August 2020 - May 2024)